

Molecular characterization of *Macrophomina phaseolina* and *Fusarium* species by a single primer RAPD technique

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Abstract

Charcoal root rot and wilt, are two economically important diseases of many crop plants in North and South America, Asia and Africa and some parts of Europe. Genetic variation in 43 isolates of *Macrophomina phaseolina* and 22 isolates of *Fusarium* species, collected from geographically distinct regions over a range of hosts, was studied using random amplified polymorphic DNA (RAPD) markers. Initially, 210 arbitrary nucleotide (10-mer) primers were tested for amplification of genomic DNA of one *M. phaseolina* isolate, 70 primers amplified the genomic DNA of *M. phaseolina*. One primer OPA-13 (5'-CAGCACCCAC-3') produced fingerprint profiles, which clearly distinguished between the different isolates of *M. phaseolina*. UPGMA analysis classified these isolates into five major groups. By primer OPA-13, 22 isolates of pathogenic and non-pathogenic *Fusarium* species of different *formae-speciales* and races, were also distinguished from *M. phaseolina*. This marker is useful for distinguishing between these two important plant pathogens irrespective of hosts, virulence spectrum and races. This is the first report of reliable diagnosis of two soilborne pathogens (root/collar rot and wilt causing pathogens) at the level of isolates, *formae-speciales* and races by a single primer RAPD procedure with uniform PCR conditions.

Key words: *Macrophomina phaseolina* – *Fusarium oxysporum* – genomic fingerprinting – single arbitrary primer

Introduction

The charcoal root rot fungus *Macrophomina phaseolina* (Tassi) Goid, has a wide host range, infecting about 500 plant species in more than 100 families throughout the world (Mihail and Taylor 1995; Srivastava *et al.* 2001). The fungus causes charcoal rot of many crops including bean, sorghum, cotton, soybean and corn under favorable conditions (Mihail 1992). Charcoal root rot (CRR) is an economically important disease of many crop plants in North and South America, Asia and Africa and some parts of Europe. Soybean yield loss due to CRR in Argentina, Brazil, Canada, India, Paraguay and United States during 1994 was 1.23 million metric tons (Wrather 1995; Wrather *et al.* 1997). In the United States alone the yield loss in soybean was 0.28 million metric tons (Wrather *et al.* 1997, 2001). Visual inspection of external symptoms on plant roots and whole plants has been the method of choice for diagnosis of charcoal/ collar rot infection of *M. phaseolina*. Visual inspection by untrained personnel may mix up charcoal rot symptoms and *Fusarium* wilt. In addition, the genus *Macrophomina* has broad species definitions, and is comprised of a single species, *phaseolina*. Many different synonyms have been ascribed to the fungus, including *M. phaseoli* (Maubl.) Ashby, *Macrophoma conchoci* Swada, *M. cajani* Syd. and Butl., *Sclerotium bataticola* Taub., and *Rhizoctonia bataticola* (Taub.) Butler (Mihail 1992). The taxonomy of *M. phaseolina* is confusing, and no reliable classification system has been proposed. The fungus is considered to belong to the Coelomycetes, the presence of a teleomorph is yet to be confirmed (Mihail 1992). Subdivisions of this monospecific genus are often based on virulence to a parti-

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cular set of differential host cultivars that vary in disease resistance. Although useful in phytopathology, these tests bear inherent problems including dependence on different host cultivars, influence of numerous variables such as cropping systems (Singh *et al.* 1990), tillage practices (Wrather *et al.* 1998), temperature, irrigation, soil water and drought stress (Kendig *et al.* 2000; Mayek-Perez *et al.* 2002). Another important problem concerning *Macrophomina* disease control is non-availability of suitable disease resistant cultivars for use in breeding (Diourte *et al.* 1995; Mayek-Perez *et al.* 1997; Smith and Carvil 1997; Lambhate *et al.* 2002).

The genera *Macrophomina* and *Fusarium* have some common hosts and tend to form mixed infections under conducive atmosphere conditions resulting in charcoal root/collar rot and wilt (Bhatti and Kraft 1992). Both species, *M. phaseolina* and *F. oxysporum*, are soil borne, infect a range of hosts at root, stem and collar region and cause cortical and vascular discoloration. Therefore, there is an urgent need to devise a diagnostic tool for the identification of these pathogens at genus, species and isolate level.

Conventional methods are impaired by a certain amount of subjectivity in quantitative evaluation of virulence, by the inability of characterizing non-pathogenic isolates of the fungus, and by the existence of physiological races (Echavez-Badel and Perdomo 1991; Sulaimen and Patil 1996). Sub-species of *M. phaseolina* were classified based on differences in microsclerotia size, cultural characteristics, chlorate sensitivity, pycnidia formation and pathogenicity (Cloud and Rupe 1991; Mihail 1992; Mihail and Taylor 1995; Karunanithi *et al.* 1999; Suriachandraselvan and Seetharaman 2000). Attempts based on molecular techniques to develop faster and unambiguous methods for the differentiation of *M. phaseolina* have been made. Among these were analyses employing RFLPs of rDNA ITS (Su *et al.* 2001), RAPD (Su *et al.* 2001) and AFLP (Vandemark *et al.* 2000). However, none of these methods have so far been able to differentiate isolates of *M. phaseolina* from specific hosts and geographical locations. Molecular markers have also been used to characterize genetic diversity of different isolates of *Fusarium oxysporum* (Edel *et al.* 1995; Kistler 1997; Alves-Santos *et al.* 1999), *formae-speciales* (Gherabay 1999) and races within form-species (Kelley *et al.* 1994; Wang *et al.* 2001; Jimenez-Gasco *et al.* 2001; Jimenez-Gasco and Jimenez-Diaz 2003).

The objectives of this study were i. to characterize genetic variability within isolates of *M. phaseolina* and different isolates of *Fusarium* species collected from different geographical regions, obtained from different hosts, and ii. identification of *Macrophomina phaseolina* and *Fusarium* by random amplified polymorphic DNA markers.

Materials and methods

Fungal isolates and culture conditions. Isolates of *M. phaseolina* were obtained from various geographical sites in India, North America and other parts of the world, cultured on Potato Dextrose Agar slants, and stored at 4°C prior to use (Table 1).

Genomic DNA extraction and PCR amplification. Pure cultures of *M. phaseolina* and *Fusarium* were grown in potato dextrose broth (PDB) for 10 days at 25–28°C in the dark. Mycelia were harvested by filtration through filter paper (Whatman No. 1). The harvested mycelia were either used immediately for DNA extraction or stored at –70°C until use. DNA was isolated using the CTAB method as described by Murray and Thompson (1980). The RAPD-PCR amplifications were carried out in a total volume of 25 µl containing 15–25 ng of genomic DNA (Williams *et al.* 1990). The reaction buffer consisted of 2.5 µl of a 10 x buffer (500 mM KCl, 15 mM MgCl₂, 100 mM Tris-HCl, pH 8.0), 0.2 µM RAPD primers Kits OPA, OPB, OPC, OPE, OPF, OPG (Opeiron Technologies Inc.), and ninety S series 10-mer primers (S1–S90) (Biobasic, Canada), 0.2 mM of each dNTP (Fermentas Inc., USA) and 0.2 µl of *Taq* polymerase (Bangalore Genei, India). The PCR amplifications were performed using a thermal cycler (Applied Biosystems) programmed for initial denaturation at 94°C for 5 min followed by 45 cycles at 94°C for 1 min, 35°C for 1 min 72°C for 2 min and a final cycle at 72°C for 7 min. All amplified DNA products were resolved by electrophoresis on agarose gel (1.4%) in 0.5 x TAE buffer, stained with ethidium bromide and photographed under UV light. All experiments were repeated at least three times.

Data analysis. DNA bands that could be scored unequivocally for presence or absence were included in the analysis. Faint bands were not scored. Binary matrices were analyzed by NTSYS-pc (version 2.0; Exeter Biological Software, Setauket, NY). Jaccard's coefficients were clustered to generate dendrograms using the SAHN clustering program, selecting the unweighted pair-group method with arithmetic average (UPGMA) algorithm in NTSYS-pc (Rohlf 1998). To determine robustness of the dendrogram, the data were bootstrapped with 2000 replications by the computer programme WINBOOT (Yap and Nelson 1996).

Results

Genetic variation was detected among 43 *M. phaseolina* isolates and 22 isolates of *Fusarium* species of different *formae-speciales* and races using RAPD markers

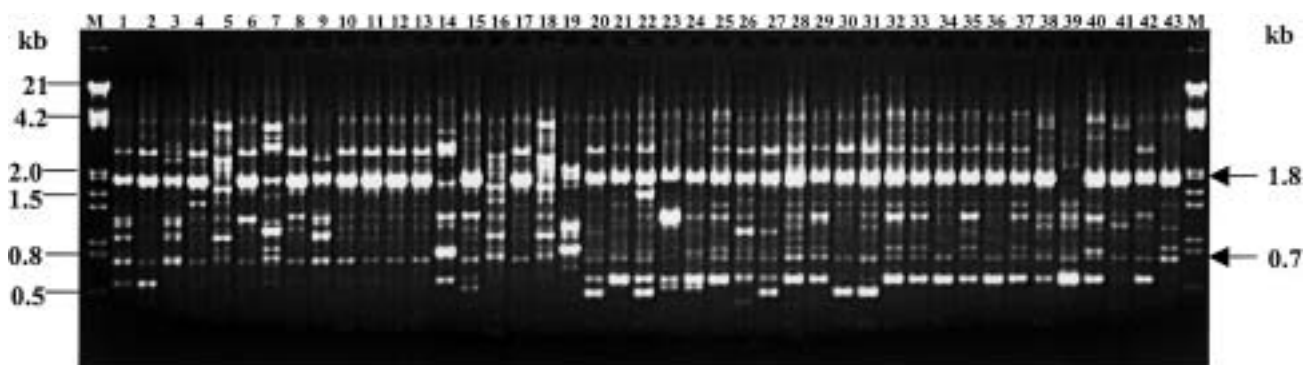


Fig. 1. RAPD profiles of different isolates of *Macrophomina phaseolina* obtained with prime OPA13. M = Molecular weight markers (Lambda DNA cut with *Eco* RI and *Hind* III) (Bangalore Genei, Bangalore, India). Lanes 1 to 43, are isolates of *M. phaseolina*. Arrows indicate the genus specific bands (1.8 and 0.7 kb) of *M. phaseolina*. PCR products were resolved on 1.4% agarose gel in TAE (0.5 X) buffer.

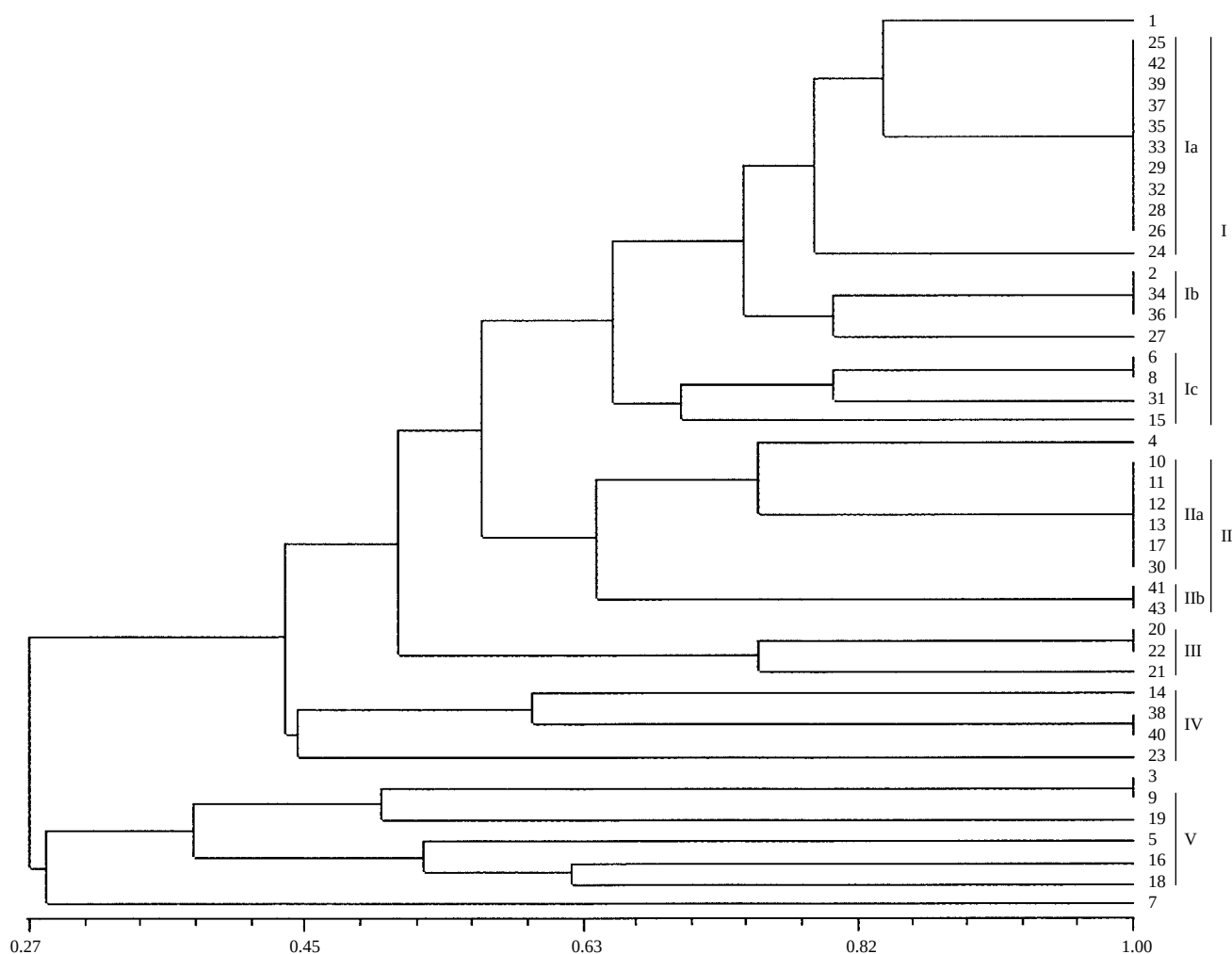


Fig. 2. Dendrogram obtained from 43 isolates of *Macrophomina phaseolina* with UPGMA based on Jaccard's coefficient. Branches are labeled by isolate numbers. The line below the dendrogram represents the similarity index.

Table 1. Source of *Macrophomina phaseolina* and *Fusarium* isolates

Name of Pathogen	Isolate no.	Isolate ID	Place of collection	Year	Host	Source
<i>Macrophomina phaseolina</i>	1	1800	New Delhi, India	1973	<i>Glycine max</i>	ITCC
	2	4203	Udaipur, India	1991	<i>Sesamum indicum</i>	ITCC
	3	1568	Hoogly, India	1970	<i>Corchorus capsularis</i>	ITCC
	4	480	Lyallpur, Pakistan	1945	<i>Glycine max</i>	ITCC
	5	481	Lyallpur, Pakistan	1945	<i>Arachis hypogaea</i>	ITCC
	6	3134	New Delhi, India	1982	<i>Cajanus cajan</i>	ITCC
	7	3083	Baghdad, Iraq	1982	not known	ITCC
	8	1569	Kalyani, India	1970	<i>Kochia trichophylla</i>	ITCC
	9	1567	Kayani, India	1970	<i>Glycine max</i>	ITCC
	10	3677	Nagpur, India	1987	<i>Gossypium barbadense</i>	ITCC
	11	3546	Assam, India	1986	<i>Phaseolus mungo</i>	ITCC
	12	49	Coimbatore, India	1934	<i>Phaseolus vulgaris</i>	ITCC
	13	482	Lyallpur, Pakistan	1945	<i>Glycine max</i>	ITCC
	14	3985	Kalyani, India	1989	<i>Psidium guajava</i>	ITCC
	15	4253	Jobner, India	1992	<i>Vigna unguiculata</i>	ITCC
	16	50	Coimbatore, India	1934	<i>Glycine hispida</i>	ITCC
	17	1705	Bhubaneswar, India	1972	<i>Corchorus capsularis</i>	ITCC
	18	249	Coimbatore, India	1940	<i>Glycine hispida</i>	ITCC
	19	CH-15	Varanasi, India	1995	<i>Cicer arietinum</i>	Jana, 1998
	20	LA B 5B	LA, USA	1998	<i>Gossypium</i> sp.	John Rupe
	21	AL C 27B	AL, USA	1999	<i>Gossypium</i> sp.	John Rupe
	22	LA 19B	LA, USA	1999	<i>Gossypium</i> sp.	John Rupe
	23	204 2R	Conwey, AR, USA	1999	<i>Glycine max</i>	John Rupe
	24	AL 9A	AL, USA	1998	<i>Gossypium</i> sp.	John Rupe
	25	AL C 4A	AL, USA	1999	<i>Gossypium</i> sp.	John Rupe
	26	TX 3 C 23B	TX, USA	1997	<i>Gossypium</i> sp.	John Rupe
	27	CA C 45B	CA, USA	1999	<i>Gossypium</i> sp.	John Rupe
	28	PD 302 5R	Conwey, USA	1999	<i>Glycine max</i>	John Rupe
	29	MS 304 2R	Greenville, MS, USA	1999	<i>Glycine max</i>	John Rupe
	30	AR 1063-13	Flood Pine Tree, AR, USA	1999	<i>Glycine max</i>	John Rupe
	31	R-WA-1	Fayetteville, AR, USA	2000	<i>Budelia</i> sp.	John Rupe
	32	105-2R	Conwey, AR, USA	1999	<i>Glycine max</i>	John Rupe
	33	1072-1	Flood Pine Tree, AR, USA	1999	<i>Glycine max</i>	John Rupe
	34	MS 306 2R	Greenville, MS, USA	1999	<i>Glycine max</i>	John Rupe
	35	PD 303 3R	Conwey, AR, USA	1999	<i>Glycine max</i>	John Rupe
	36	105 3R	Conwey, USA	1999	<i>Glycine max</i>	John Rupe
	37	2 CCC-5B	Clarkedale, AR, USA	1999	<i>Gossypium</i> sp.	John Rupe
	38	305-5R	Conwey, AR, USA	1999	<i>Glycine max</i>	John Rupe
	39	ICG-7459	Hyderabad, India	2000	<i>Cicer arietinum</i>	ICRISAT
	40	ICCG-9116	Hyderabad, India	2000	<i>Cicer arietinum</i>	ICRISAT
	41	87	Hyderabad, India	2000	<i>Cicer arietinum</i>	ICRISAT
	42	Not available	Hyderabad, India	2000	<i>Cicer arietinum</i>	ICRISAT
	43	PAU-5	Ludhiana, India	1999	<i>Abelmoschus esculentus</i>	V. K. Kaul
<i>Fusarium solani</i>	44	136	IARI, New Delhi	2000	<i>Pisum sativum</i>	ITCC
<i>F. o. f. sp. vasinfectum</i> (Fov)	45	3865	New Delhi, India	1988	<i>Gossypium</i> sp.	ITCC
<i>F. o. f. sp. pisi</i>	46	4814	Imphal, India	1998	<i>Pisum sativum</i>	ITCC
<i>F. o. f. sp. pisi</i>	47	976	Pune, India	1956	<i>Pisum sativum</i>	ITCC
<i>F. o. f. sp. pisi</i>	48	195	CBS, Netherlands	1938	<i>Pisum sativum</i>	ITCC
<i>F. o. f. sp. ciceri</i> (race-I)	49	2789	Hyderabad, India	1981	<i>Cicer arietinum</i>	ITCC
<i>F. o. f. sp. ciceri</i> (Race-II)	50	2790	Kanpur, India	1981	<i>Cicer arietinum</i>	ITCC
<i>F. o. f. sp. ciceri</i> (Race-III)	51	2791	Gurdaspur, India	1981	<i>Cicer arietinum</i>	ITCC

Table 1. Continued

Name of Pathogen	Isolate no.	Isolate ID	Place of collection	Year	Host	Source
<i>F. o. f. sp. carthami</i>	52	AK-1	Akola, Maharastra, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	53	–	Annigeri, Karnataka, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	54	–	Solapur, Mahasrastra, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	55	AK-2	Akola, Maharastra, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	56	–	Tandur, Andhra Pradesh, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	57	–	Phaltan, Maharastra, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. o. f. sp. carthami</i>	58	–	DRR, Hyderabad, India	2002	<i>Carthmus tinctorius</i>	R. D. Prasad
<i>F. oxysporum</i>	59	179	CBS, Netherland	1938	Not known	ITCC
<i>F. oxysporum</i>	60	564	CBS, Netherland	1946	Not known	ITCC
<i>F. oxysporum</i>	61	563	CBS, Netherland	1946	Not known	ITCC
<i>F. oxysporum</i>	62	5247	Madurai, India	2001	<i>Eichornia crassipes</i>	ITCC
<i>F. moniliforme</i>	63	954	Anand, India	1955	<i>Cyamopsis sp.</i>	ITCC
<i>F. o. f. sp. subglutinans</i>	64	4019	Hyderabad, India	1990	<i>Carthamus tinctorius</i>	ITCC
<i>F. graminearum</i>	65	3437	Wellington, Tamil Nadu, India	1985	<i>Triticum aestivum</i>	ITCC

ITCC = Indian Type Culture Collection; ICRISAT = International Crop Research Institute of Semi Arid Tropics

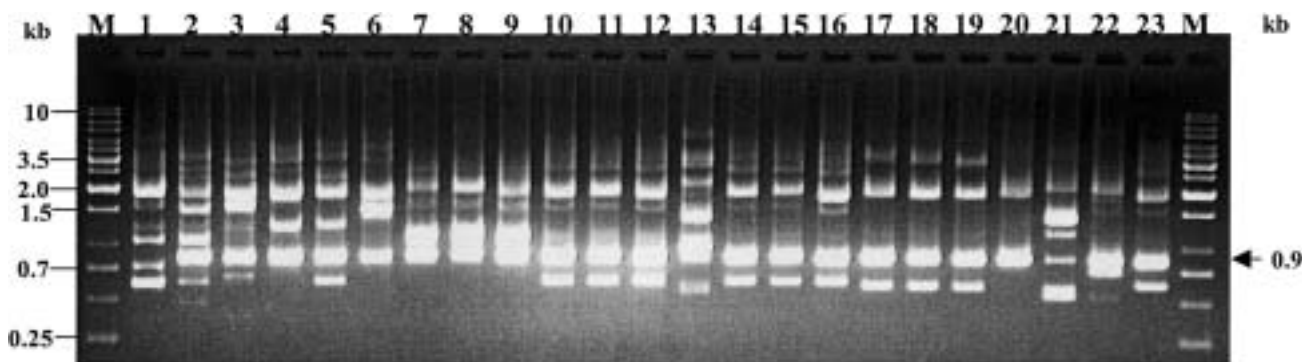


Fig. 3. RAPD profiles of one isolate of *Macrophomina* and 22 isolates of *Fusarium* species obtained with primer OPA-13. Lane 1: *Macrophomina* isolate and lanes 2–23: isolates of *Fusarium* species. M: 1 kb molecular weight markers (Farmentas Inc., MD, USA). The arrow indicates the genus specific DNA band of *Fusarium*. PCR products were resolved on 1.4% agarose gels.

(Table 1). Among the 210 primers, only 70 amplified the genomic DNA of one *M. phaseolina* (isolate no. 1), and produced bands ranging from 0.3 to 3.5 kb. The primer OPA-13 amplified the genomic DNA of all the isolates of *M. phaseolina* producing 2–8 bands of sizes between 0.3 and 3.5 kb. Nearly all isolates of *M. phaseolina*, irrespective of their host and geographical origin, exhibited a common band of 1.8 kb, except isolates 5, 14, 16, 18 and 39, and 0.7 kb (exceptions: isolate 14 and 19; Fig. 1). Primer OPA-13 shows potential for discriminating between the different isolates of *M. phaseolina* (Fig. 1).

Data were analyzed with UPGMA by using NTSYS and a dendrogram was constructed (Fig. 2). The resulting dendrogram showed relatedness and genetic distances among different isolates obtained from different hosts and geographical regions. Genetic similarity coefficients ranged from 0.27 to 1 (Fig. 2). Five different major groups were obtained among the isolates. The first group consists of 20 isolates, which include 3 sub-groups (Ia, Ib, Ic). Sub-group Ia consists of 11 isolates, of which 9 isolates (isolate no. 25, 26, 28, 29, 32, 33, 35, 37, 39) show 100% genetic similarity. All of these iso-

lates were obtained from USA, except isolate no. 42, which was collected in India. Subgroup 1b consists of 3 isolates, while two isolates (no. 6 and 8) were grouped together in subgroup 1c, exhibiting a high bootstrap value of 69%. A second group consists of 9 isolates, with two sub-groups IIa and IIb (Fig. 2). Similarly, isolates obtained from the USA formed a distinct group (group III; Fig. 2). All isolates collected from India grouped consistently in group IV and V. Isolate no. 7 from Iraq did not fit in any of the groups.

The primer OPA-13 was also evaluated for the characterization of 22 *Fusarium* isolates, form-species and races (Table 1). Different isolates of *Fusarium* species, irrespective of form-species and race, were amplified with this primer (Fig. 3). A band of 0.9 kb was amplified by OPA-13. This band was consistently present in all

isolates. It separated *Fusarium* isolates from those of *Macrophomina*. A 1 kb band, amplified from three isolates of *F. oxysporum* f.sp. *ciceri*, was absent in other *Fusarium* species. Similarly, a 0.6 kb size band was present in all *F. oxysporum* f. sp. *carthami* isolates, whereas it was absent in all other *Fusarium* species (Fig. 3). The dendrogram consists of 23 isolates; the *M. phaseolina* isolate did not join any of the *Fusarium* groups (Fig. 4). The bootstrap values showed clear differences among different groups of *Fusarium* species, which were further classified into four major groups. Different species and form-species of *Fusarium* formed group I with 100% genetic similarity (Fig. 4). Group II is exclusively formed by isolates of *F. oxysporum* f. sp. *carthami*, exhibiting 100% genetic similarity (bootstrap value of 38.6%). Group III is represented by 2 isolates of *F.*

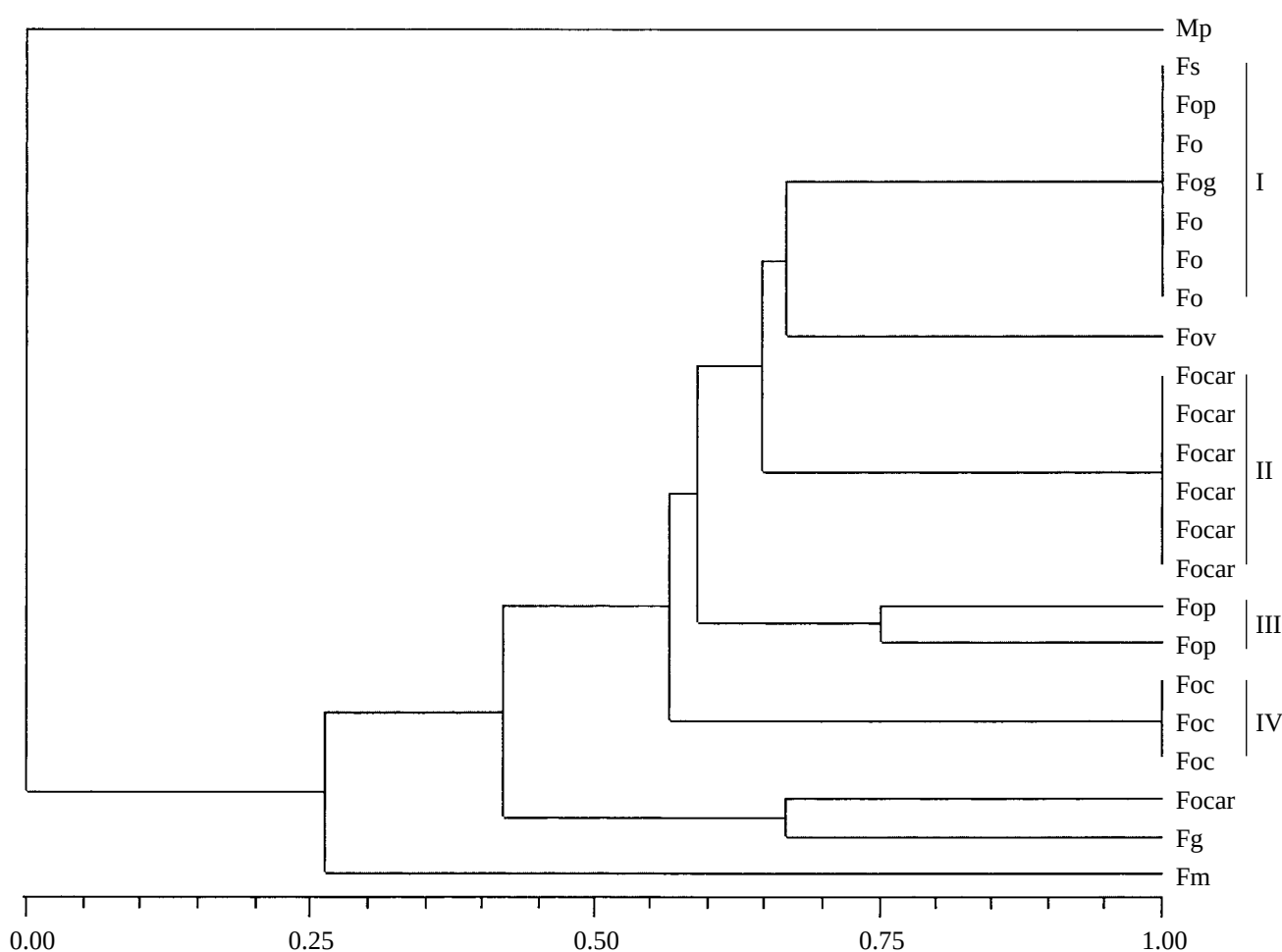


Fig. 4. Dendrogram obtained from 22 isolates of *Fusarium* species with UPGMA based on Jaccard's coefficient. Isolates of different species are indicated at the termini of branches. Mp = *Macrophomina phaseolina*, Fs = *Fusarium solani*, Fog = *F. o. f. sp. subglutinans*, Fo = *F. oxysporum*, Focar = *F. o. f. sp. carthami*, Fop = *F. o. f. sp. pisi*, Fov = *F. o. f. sp. vasinfectum*, Foc = *F. o. f. sp. ciceri*, Fg = *F. graminearum*, Fm = *F. moniliforme*. The line below the dendrogram represents the similarity index.

oxysporum f. sp. *pisi* with 75% genetic similarity (bootstrap value of 53.9%). Group IV is formed exclusively from *F. oxysporum* f. sp. *ciceri* with 100% genetic similarity and 65.3% bootstrap value (Fig. 4).

Discussion

Our results showed that the single RAPD primer OPA-13 can be used to identify and discriminate several isolates of *M. phaseolina* and *Fusarium* species obtained from 20 hosts including soybean, cotton, chickpea, pea and safflower. The fingerprinting pattern enabled us to distinguish between the isolates of *M. phaseolina* from different hosts. This could be useful as a taxonomic marker for specific population studies. Genetic variability of 43 isolates of *M. phaseolina* was detected on the basis of their amplification pattern. Molecular techniques using RFLP, RAPD, AFLP, microsatellite, mini-satellite fingerprinting have been used to identify and differentiate isolates of several filamentous fungi at population, genus and species level (Carter *et al.* 2003; Sharma *et al.* 1999; 2003). However, there are only few reports on the molecular characterization of *M. phaseolina* isolates in relation to host specificity (Vandemark *et al.* 2000; Su *et al.* 2001). These studies were confining to a particular geographical area of the United States. The genetic differences among the isolates of *M. phaseolina* may possibly be due to their isolation from diverse hosts and distinct geographical regions. Although, the isolates of *M. phaseolina*, we used in this study, showed morphological similarities (Jana 1998), they exhibited a wide range of genetic variability with RAPD markers. With a few isolates Su *et al.* (2001) have shown genetic relatedness of *M. phaseolina* according to their host specialization. Since RFLP is labour intensive and requires several restriction enzymes, processing steps, and high amounts of DNA, RAPD analysis was used during this study, which is a relatively simple and rapid technique. General assessments of the technique and discussions of its applicational range have been reviewed (Schäfer and Wöstemeyer 1992; Wöstemeyer *et al.* 1992; Harris 1999).

Analysis of band patterns obtained with primer OPA-13 showed that *M. phaseolina* obtained from different geographical locations are closely related to each other despite of host differences. The analysis revealed five groups. Similarly, isolates from different species, form-species and races of the genus *Fusarium* were classified into four groups according to their host.

Several molecular markers and DNA based probes were developed for detecting *Fusarium* species and its different races in relation to specific hosts (Kelly *et al.* 1994; Kistler 1997; Gherbawy 1999; Wang *et al.* 2001,

Alves-Santos *et al.* 1999, 2002; Jimenez-Gasco *et al.* 2001; Jimenez-Gasco and Jimenez-Diaz 2003). However, there are no reports for using molecular markers for the identification of isolates of *M. phaseolina* and *Fusarium* species in mixed infection complexes. In such conditions diagnosis of root rot and wilt is more complicated by cultural and serological methods (Banks *et al.* 1996; Sriavastava and Arora 1997). This study also revealed that a single primer can clearly distinguish the two genera *Macrophomina* and *Fusarium* at the molecular level. The single primer method, at least at the genus level circumvents trial and error methods or the use of different primers to detect and discriminate the two important pathogens.

The common RAPD bands, shared between these two genera, may allow the development of genus specific DNA markers for *Macrophomina* and *Fusarium*, which will facilitate identification of these two pathogens by multiplex analysis. This will be a valuable tool for epidemiological studies of these fungi and may be important for the development of selective pathogen management strategies.

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